

to play the most important role in reshaping the air transport industry in 2019 are:

Biometrics. Biometrics-enabled self-service boarding gates, where your face becomes your passport, have been implemented at the entrances of many airline lounges. Integration of this technology into signages and flight information display systems is used to serve passengers with personalised information and offers.

Biometric technology can also be used onboard to facilitate seatback immigration and simplify in-flight payments. Airlines, airports and their partners are continuously working towards the goal of creating a biometrics-enabled single passenger token to offer effortless journeys.

Biometric technology such as fingerprint scanning or facial recognition on smartphones has also been adopted by airlines and airports to transform the customer journey. Fingerprint scanning for identification purposes to check in baggage could eliminate the need for boarding passes and cut down on the need for crew to be present to manually scan them at the gate.

Blockchain. Blockchain technology can eliminate the need to rely on physical ID proofs by saving passengers' data maintained in a virtual de-centralised database. It can be extremely useful in building a robust security system for managing customer data by using a cloud-based document management solution. This will save time, money and resources in the long run, helping airlines become more efficient, safer and compliant. This data can also be used in increasing the operational efficiencies through predictive analytics.

Artificial intelligence. Chatbots can be used to answer simple questions, freeing up resources (human agents) for more complex customer service issues. Artificial intelligence (AI) also holds potential in a number of other areas, such as real-time predictive pricing, predictive aircraft maintenance and operational efficiency on the airfield.

Travel experiences can be daunt-

ing when you are in an unfamiliar environment and do not speak the local language. This could be solved by AI and machine learning (ML) technologies with translation apps and devices.

Robotics. Robots can assist in various tasks from transporting luggage to proactively identifying potential security risks. Customer-facing robots can provide on-the-spot assistance to passengers like customer management, car parking and so on. Baggage management can also be improved with re-usable electronic bag tags and door-to-door pick-and-drop facility.

Recent advancements in sensors and condition monitoring technologies have led to continuous data collection and evaluation, significantly minimising the number and cost of unscheduled maintenance

Beacon technology for indoor positioning systems. Beacons can make navigation easy for travellers between different terminals at the airport. It can also help to know where passengers are and send them personalised and relevant information accordingly. This information can be about boarding gate number, baggage or flight status, or about the shops around.

AR and VR. Augmented reality (AR) allows passengers to use the camera on their device to view AR directions to wherever they need to go within the terminal. It helps them navigate the complex layout of airports.

VR-based apps can be used for training new crew members and engineers about engine mechanics.

Cybersecurity and cloud. Everything that airlines do to improve their digital touchpoints depend on the security of their IT systems. While technology leaders and industry stakeholders are working on a

number of solutions, airlines should keep cybersecurity at the top of their priority list.

The IoT. Sensors can detect and report faults to maintenance teams on the ground, removing the need for the crew to manually do so. Addition of sensors to aircraft seats will enable the crew to monitor individual passenger health and well-being. Sensors such as hydraulic pressure and brake temperature provide real-time data to help predict early malfunctions or diagnose the remaining life cycle of the landing gear. Researchers are developing resilient ML-based autopilot systems that can adapt to changing conditions.

Airlines are using mobile platforms to connect with their customers throughout the journey, starting from booking of a flight to deplaning it. Deployment of connected objects can transform the manufacturing process with new tools and services that generate a drop in production costs and increase productivity. Overall, it can reduce both time and cost of maintenance.

Conclusion

With the rise of Industry 4.0, railway companies can now ensure that they are prepared to avoid the surprise of equipment downtime. The railway industry is on its way to integrate predictive maintenance and Big Data. Recent advancements in sensors and condition monitoring technologies have led to continuous data collection and evaluation, significantly minimising the number and cost of unscheduled maintenance.

Most significant improvements have been evidenced by more informative and user-friendly websites, mobile applications for real-time information about vehicles in motion, and e-ticket purchases and timetable information implemented at stations and stops. Airline technology is likely to evolve over time to the point where one crew member can handle a sizable chunk of a flight given an anticipated pilot shortage in the future. With automation a remote pilot on the ground will be able to take the second seat on a flight or flight segment. **EFY**